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It undergoes several mitotic division to form embryo. The embryo develops into a mature sporophytic plant.

In the life cycle of *Selaginella* alternation of sporophytic and gametophytic generation is present (Figure 2.30).

2. 5.5 Adiantum

Division – Pteropsida
Class - Leptosporangiopsida
Order – Filicales
Family – Polypodiaceae
Genus – *Adiantum*

Adiantum is commonly known as 'Maiden hair fern' or 'Walking fern'. They are distributed in the tropical and temperate regions of the world. Some of the Indian species include *Adiantum capillus-veneris, Adiantum pedatum, Adiantum caudatum* and *Adiantum venustum*. The sporophyte is differentiated into rhizome, roots and leaves Figure 2.31.

External features

Rhizome

The rhizome is a perennial, subterranean dichotomously branched structure and is creeping in *A. capillus-veneris* or may be erect as in *A. caudatum*. It is covered with persistent leaf bases and hairy outgrowths called **ramenta**.

Root

The roots are adventitious and arise from the rhizome.

Leaf

The leaves are also called fronds and are pinnately compound (unipinnate- *A. caudatum*, bipinnate- *A. capillus-veneris*) the young leaves are circinately coiled. The petiole is long, black and shiny. The venation is free and dichotomous in all the species. The vein spread in a fan-like manner in the lamina. The leaves bear marginal sori which are covered by a **false indusium.**

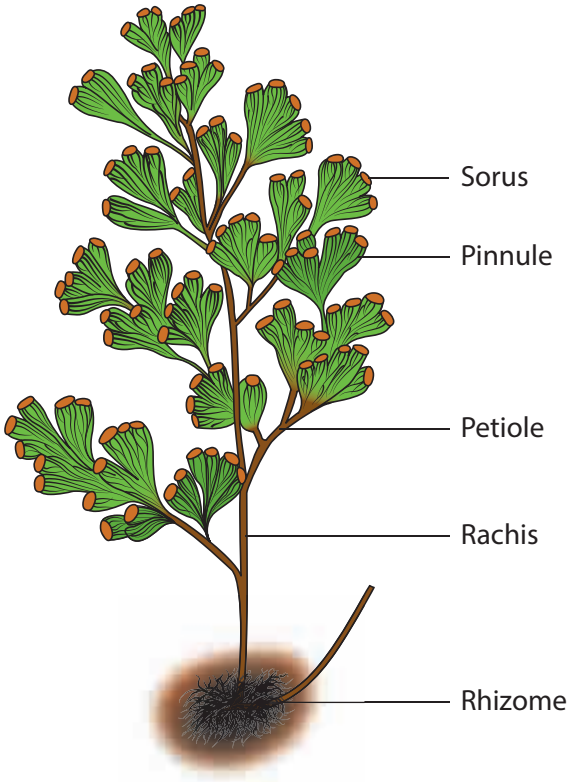


Figure 2.31: Adiantum Habit

Internal structure

Root

The root is differentiated into epidermis, cortex and central vascular cylinder.

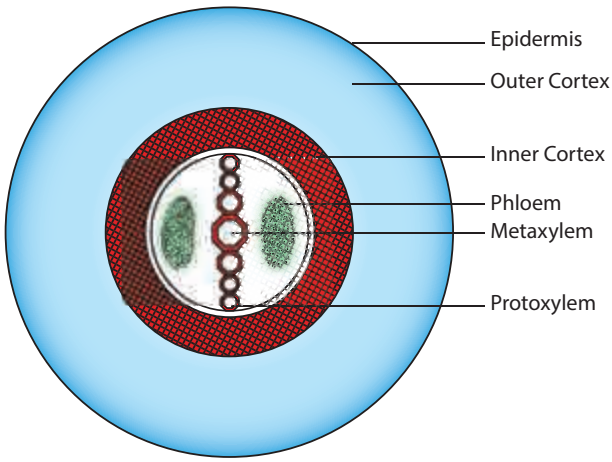


Figure 2.32: T.S. of root